

addresses would have been exhausted and therefore the complete adoption of IPv6 worldwide is only a matter of time, and critical to the success of IoT. IPv6 creates an independent network parallel to that of IPv4. However, intermediate trans-protocol systems are available and are indeed widely used today as IPv6 is gradually adopted. The numbers involved here are ridiculous and, for those who are analytically inclined, very exciting.

Currently, there are over three billion internet users. Imagine each of those individuals using basic electronic equipment at home, work and in their cars. Now imagine each of those devices generating small chunks of information throughout the day – when you switch on your TV, when you put on some music, when you ask the in-car navigation system for directions to the local mall, when you make a phone call – and you soon begin to see that an enormous amount of data is generated by people all over the world, and how these tiny bits start adding up.

According to IC Insights, a market research analysis firm focussed on the semiconductor industry, global sales from IoT are expected to grow from \$57.7 billion in 2015 to an astounding \$103.6 billion in 2018, representing a compound annual growth rate of 21% over five years starting from 2013.

3. The improvement

Convenience is usually a commodity that is worth its price, and IoT (at the cost of making us more lazy) has immense potential to bring about transformative convenience in our lives – both personal and public.

Home automation: Numerous companies now manufacture “smart plugs” that allow you to control anything plugged into a power socket using just an app. By using multiple smart plugs, you could control your entire home remotely.

Industrial applications: Sensors on manufacturing equipment can monitor their designated thresholds, and error reports can automatically be sent to the manufacturers. This helps early prediction of equipment malfunction and therefore improves service maintenance.

Logistics: UPS, one of the world’s largest shipping companies, uses analytics on large-scale data obtained from hundreds of data points in each of their 80,000 delivery vehicles. This has helped UPS optimise its delivery process and reduce fuel costs and idling time.

Smart cities: Smart parking meters in Barcelona, Spain are connected to the city-wide Wi-Fi network. This system gives out real-time updates about available parking spaces, and also allows the residents to pay through